Discovering adsorption rate laws from column data fails long before the rate constant is lost

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Abstract

Equation-discovery methods promise to learn the uptake rate law of an adsorption column from its data, but where they can is unknown. In a pre-registered computational study spanning four decades of Damköhler number (Da), three Péclet numbers, noise and isotherm error, we compared the Da at which the best discovery method stopped recovering the linear-driving-force law with the Da at which its rate constant stopped being practically identifiable. In all 30 conditions where both boundaries existed, discovery failed first, by 1.6 to 3.4 decades. An isotherm error of 5 % prevented discovery at every rate, while the rate constant remained identifiable from interior probes. Weak-form sparse regression worked at low Da and symbolic regression in a window near Da 1 to 10; neither reached the range where the rate constant stayed identifiable. Column data carry information about the rate long after its law can no longer be discovered.

A fixed-bed adsorption experiment reports concentrations and temperatures over time, most often at the column outlet. What an engineer needs from it is the uptake rate law, usually written as the linear driving force (LDF), $\partial q/\partial t = k\,(q^*(c,T) - q)$. Equation-discovery methods offer to find the law itself, not only its coefficient: sparse regression over a library of candidate terms [1, 2] and its weak-form [3, 4], ensemble [5] and exact best-subset [6] variants, symbolic regression [7, 8], and neural networks embedded in the governing equations [9]. For adsorption, one study has done this from column data: Santana et al. trained a neural uptake term on noisy outlet breakthrough curves and read the LDF form back out, with the isotherm given exactly and at one rate and one noise level [9]. Repeated database searches found no other study that discovers an adsorption rate law from column data. Where such a result holds, and where it stops holding, is not known.

Part of the answer is classical. Near local equilibrium the mass-transfer coefficient is poorly determined by column data [10]: finite rates then act only through an apparent dispersion [11], the LDF coefficient can be extracted wrongly from outlet breakthrough [12], and most of the output variability belongs to the isotherm [13, 14]. Yet in the constant-pattern regime a lumped rate coefficient is recovered from breakthrough consistently with theory [15], so any limit must be conditioned on dispersion, isotherm shape, noise and isotherm error. Equation discovery has its

own limits. On a single transient trajectory that settles onto a slow manifold, as a breakthrough run does, the candidate terms become nearly collinear and the data barely distinguish them [16–18]; noise induces phase transitions in what can be recovered [19, 20]; and unique recovery needs conditions that non-chaotic dynamics generally lack [21]. These results say that discovery must fail somewhere for column data. None says where, for which methods, or whether it fails where the rate constant itself stops being identifiable.

The question matters most where the law is not the textbook one. In the water-harvesting framework MOF-303 the water diffusion rate passes through a minimum at the steep step of the isotherm, and the LDF approximation is least accurate there [22, 23]; a loading-dependent rate has been fitted by hand [24]. A method that could discover such a law from column data would be useful. Whether one can, at the noise and isotherm accuracy of a real experiment, is what this study tests.

We asked, in a pre-registered computational study, for which regimes the adsorption rate law can be discovered from data a column experiment can produce, and whether discovery fails at the same boundary as the practical identifiability of the rate constant [25], or earlier. Ground truth came from a validated non-isothermal column model [26], for a Langmuir sorbent and a MOF-303-like step isotherm, over four decades of Damköhler number, three Péclet numbers, four noise levels and four levels of isotherm error, observed through interior probes of concentration, uptake and temperature, through probes without uptake, or at the outlet alone. Methods from each family above were run in their strongest available configuration, and identifiability was judged by profile likelihood on the same observations. The design, thresholds and verdict words were frozen before any method ran on the grid. Discovery failed first. In every slice of the design where both boundaries could be measured, the law stopped being discovered 1.6 to 3.4 decades of Damköhler number before its rate constant stopped being identifiable. An isotherm error of 5 % prevented discovery at every rate, and one of 2 % in most conditions, while k remained identifiable from interior probes and, where tested, from the outlet alone.

Results

Both isotherms pass the positive control

At the lowest Damköhler number, with no noise and no isotherm error, discovery and identifiability must both succeed in at least 90 % of the estimation replicates, or that isotherm is withheld from every verdict. Both isotherms passed: the best method recovered the LDF law in 135 of 135 replicates, and k was identifiable in 135 of 135, for the Langmuir and for the step isotherm alike. (As first run, the step isotherm failed this check because of a search-range error in the identifiability code; the correction and both sets of verdicts are given in Supplementary Note 2.)

Discovery fails long before identifiability

H2b compares, in each slice of the design (one isotherm, one Péclet number, one noise level and one isotherm error), the Damköhler number at which the best discovery method falls to 50 % success, $\mathrm{Da}^*_{\mathrm{disc}}$, with the number at which k stops being identifiable in half the replicates, $\mathrm{Da}^*_{\mathrm{ident}}$. Of the 96 slices, 30 had both boundaries on the grid. In all 30 the verdict was "*discovery fails before identifiability*", so H2b's pre-registered overall word is the same (Fig. 2, Table 1). The gap was large everywhere: from 1.6 to 3.4 decades of Da, with a median of 2.3 decades for the Langmuir sorbent (20 slices) and 2.5 for the step isotherm (10 slices). Discovery failed between Da = 0.11 and 7.2, while k stayed identifiable up to Da between 43 and 562. In every one of the 30 slices the best discovery method was the same, exact best-subset selection in the weak form.

Noise moved both boundaries, but not equally. With 0.5 % noise and an exact isotherm, discovery held to Da ≈ 5 to 7 and identifiability to between the two highest levels (Da = 316 and 1000). At 2 % noise the discovery boundary fell to Da ≈ 1, and at 5 % to Da ≈ 0.1 to 0.3, near the bottom of the grid, while the identifiability boundary stayed between Da ≈ 57 and 172. A small error in the isotherm hurt discovery more than noise did: at 0.5 % noise, an isotherm error of 0.5 % moved $\mathrm{Da}^*_{\mathrm{disc}}$ from about 6 to about 0.2, more than a decade, while $\mathrm{Da}^*_{\mathrm{ident}}$ moved from 562 to about 360.

The 66 slices without a verdict are not missing data. In each, one of the two probabilities had no 50 % crossing on the grid, and the pre-registered words for that case are "*no boundary on the grid for one of the two*". A descriptive check made after the verdicts were seen sorts them by the boundary that is missing. The discovery boundary was missing in 59 of them. In 40, discovery never reached 50 % at any Da: all 24 slices with 5 % isotherm error, 15 of the 24 with 2 %, and one with 0.5 %. In 8, all without noise, it never fell below 50 %. In the other 11, success crossed 50 % more than once, so no single logistic crossing, the pre-registered boundary, describes it (Fig. 2). The clearest case is symbolic regression (M7) at Pe ≈ 94 with 2 % isotherm error, the best method in four slices: its success peaked at 82 to 96 % of replicates near Da = 3, in a window between Da = 1 and 10, and was zero below Da = 1 and above Da = 32. In the remaining 7 slices only the identifiability boundary was missing, because k stayed identifiable at every level or its identifiability dipped and recovered at the top of the grid (see below). Many identifiability boundaries fell exactly between two adjacent levels of Da: in every such slice, k was identifiable in all replicates at one level and in none at the next. These boundaries are reported as lying between those levels (Methods).

The identifiability boundaries at the top of the grid need one caution, found in a second descriptive check made after the verdicts were seen. With exact data the O1 profile recovered k to within 11 % at every Da up to 100, but at Da = 316 and 1000 its estimate was off by a factor of 0.6 to 24, because the adsorption front crosses a probe in only a few of the 200 observation times. Identifiability there is limited by the time resolution of the data as well as by their information content. This does not affect H2b: in all 30 slices with a verdict, k was identifiable in at least 95 % of replicates at every Da ≤ 31.6, and the highest Da ≤ 100 at which it was identifiable in half the replicates lay at least 1.1 decades above the discovery boundary.

The verdict does not depend on how success is scored. When success also required the fitted k

to lie within a factor of two of the truth (a pre-registered sensitivity analysis), the overall word and all 30 slice verdicts were unchanged.

No single error ratio predicts the boundary, and errors-in-variables could not be tested

H2a asked whether one dimensionless ratio, the realised driving force against the combined error of the observations and of the isotherm ($R = \delta/\varepsilon_{\mathrm{eff}}$), places the discovery boundary at the same value R^* in every condition. A verdict needs at least two conditions whose success crosses 50 % in R. Only one did (the step isotherm at Pe ≈ 19, observed through all probes, with $R^* = 123$, 95 % interval 113 to 135), so H2a has "*no verdict*". In the other eleven conditions success was not a monotone function of R: on probes with uptake (O1) it reached 51 to 71 % at single levels of R without a 50 % crossing of the fitted curve, and on probes without uptake (O2) it never exceeded 20 % at any level of R.

H2c asked whether an errors-in-variables estimator (M5), which treats the measured isotherm as a regressor with error, moves the discovery boundary upward. It is tested at 2 % noise on probes with uptake, in the strong form, against the best strong-form method that is not errors-in-variables, in 18 slices (three isotherm errors from 0.5 to 5 %, two isotherms, three Péclet numbers). It holds at an isotherm error if the upward shift is at least 0.25 decade in two thirds of that error's slices with a verdict, and a slice has a verdict only if both estimators have a boundary on the grid. No slice had both, so H2c has "*no verdict*" at every isotherm error and is neither supported nor falsified. The reason is the form of the regression. Errors-in-variables fitting needs the time derivative of the uptake, and in 16 of the 18 slices the comparator was strong-form best-subset selection, which, like the other strong-form sparse methods, succeeded in no replicate at any Da at this noise level; the errors-in-variables estimator reached at most 40 % (step isotherm, 5 % isotherm error), never 50 %. In the other two slices (Pe ≈ 94, 2 % isotherm error) the comparator was symbolic regression, whose success rose above 50 % and fell again within the grid, so it had no single boundary either. Because the refined grid that H2c would have used is placed around discovery boundaries, and there were none, it was not solved.

Which methods discover the law, and where

Exact best-subset selection in the weak form (M4) was the most reliable method: its success crossed 50 % within the grid in 36 of the 96 slices, against 8 for weak-form sequential thresholding (M2) and 7 for the weak-form ensemble (M3) (Fig. 4c). No strong-form sparse method had a crossing in any slice. Symbolic regression (M7), run only at Pe ≈ 94 with noise of 0.5 or 2 % and isotherm error of 0 or 2 %, behaved differently from all the sparse methods. It failed at the lowest rates, where weak-form sparse regression worked best, and succeeded at intermediate rates: up to Da ≈ 32 without isotherm error, and between Da = 1 and 10 with 2 % isotherm error, where no sparse method reached 50 % (Fig. 4a,b). The two families therefore cover different ranges of Da, and neither covers the range in which k remains identifiable. The neural-network pipeline of the closest prior work

(M1), run as a declared demonstration at one noise level, recovered the law in one of its nine fits, at Da = 3.16. Slow-manifold discovery (M9), which targets the isotherm rather than the rate law, recovered its own target only at the highest rate and without noise (67 % of replicates at Da = 1000, pooled over isotherms and Péclet numbers), and in at most 17 % elsewhere.

Without probes of uptake, k stays identifiable but the law is not discovered

Real columns are rarely fitted with probes inside the bed. With concentration and temperature probes but no uptake (O2), uptake must first be reconstructed from the gas-phase mass balance. Discovery failed on O2 everywhere: no slice had a 50 % boundary, and no method succeeded in more than 29 % of the replicates at any level of any slice. Identifiability, computed on the subgrid where every point of the profile is a full column solve (Pe ≈ 94, noise 0.5 and 2 %, isotherm error 0 and 2 %), held to Da ≈ 57 to 286 from probes without uptake, and to Da ≈ 38 to 172 from the outlet alone (O3). At the same noise and isotherm error, discovery from probes of c, q and T, a far richer observation, last reached 50 % success at Da ≈ 1 to 7 without isotherm error, and at Da ≈ 3 to 10 with 2 % isotherm error, where only symbolic regression succeeded. In these slices the outlet curve still determined k 1.3 to 2.0 decades of Da beyond the last discovered level without isotherm error, and 0.6 to 1.3 decades with 2 % isotherm error.

A loading-dependent rate is found almost only at one rate

For the step isotherm the study also simulated a rate that falls to 20 % of its value across the isotherm step (L2), a form motivated by the measurements in MOF-303. This part is exploratory. The state dependence was discovered, by the pre-registered rule, almost only at Da = 1: there the best method succeeded in 77 to 100 % of replicates at every noise level and isotherm error, against at most 58 % at any other level and none at Da ≤ 0.32 without noise. These are the best of all methods and Péclet numbers at each level, so they overstate what a single method achieves. The failure at low Da is not because the loading never reaches the step: at Da = 0.1 and 0.32 the loading rose to 41 to 47 % of capacity, past the step midpoint at 36 %, and stayed near the step for most of the run. Why the state dependence is recovered at Da = 1 and not at its neighbours was not resolved here.

Discussion

The rate law of a fixed-bed column could be discovered only over a narrow part of the range in which its rate constant could be measured. In every slice of the design where both boundaries existed, discovery failed between 1.6 and 3.4 decades of Damköhler number before identifiability did. With an isotherm error of 5 % no method discovered the law at any rate, and with 2 % the best method reached 50 % at any level in only 8 of 24 slices: at Da ≈ 0.3 under 5 % noise, or, for symbolic regression, in a window between Da = 1 and 10. For practice this sets an order of operations. Where the isotherm is known almost exactly (an error of 0.5 % already confined discovery to Da ≤ 0.2)

and the uptake is slow compared with the passage of the front (Da of order one or below), sparse regression in the weak form can propose the law. Beyond that, the law has to be assumed or chosen among a few candidates by model comparison, and the data are then good for its coefficient over a further one and a half to three decades of Da from interior probes, and, at the noise and isotherm errors tested, over roughly one more decade from the outlet alone.

Why is discovery so much more fragile? The two tasks ask different things of the same data. Identifiability asks whether a known law with one free coefficient fits well only near the true value. Discovery asks the data to choose that law among many, and, in the success rule used here, to return the coefficients of q^*_{meas} and q in the ratio -1 within 10 %. An isotherm error enters discovery as an error in a regressor, which biases exactly that ratio. In the identifiability fit the same error is absorbed into k, and the estimate stays within a factor of two of the truth. The observation that an isotherm error of 0.5 % moved the discovery boundary by more than a decade, while it moved the identifiability boundary by about 0.2 decade, is consistent with this reading, although the study was not designed to separate the mechanisms. A second effect is classical: as Da grows, uptake approaches equilibrium, the driving force $q^* - q$ shrinks and the candidate terms become nearly collinear on the slow manifold [17, 18], so the selection step loses contrast before the fit of a single coefficient does.

The discovery methods were not interchangeable. Exact best-subset selection in the weak form was the most reliable, and strong-form sparse methods, which differentiate noisy data, never produced a boundary. Symbolic regression failed where weak-form regression did best, at the lowest rates, and succeeded at intermediate Da, up to about 32 without isotherm error and between 1 and 10 with 2 % isotherm error. A combination of the two families might widen the range, but it was not tested, and neither family reached the range where k stayed identifiable. The errors-in-variables estimator could not be judged, because its strong-form comparator never had a boundary at the noise levels where it was tested; whether errors-in-variables fitting helps a weak-form search remains open.

The closest prior work recovered the LDF law from noisy outlet curves with a neural network in the governing equations, with the isotherm given exactly, at one rate [9]. Our results place that success at one end of a wider map: with an exact isotherm, discovery from interior probes survives noise of 2 to 5 % at low Da, but outside that corner it fails, and from probes without uptake no method reached 50 % at any rate. From the outlet, only our demonstration of the same neural pipeline (M1) was run; limited by computation to one noise level, one replicate per rate and fifty optimisation steps, it recovered the law in one of nine fits. For water in MOF-303, where the rate depends on loading at the isotherm step [22, 23], a loading-dependent law was recovered almost only at $\mathrm{Da} = 1$; such a law can be found, but not yet reliably.

The study has limits that bound these conclusions. All data were simulated by one validated column model with the LDF law as the truth, so the boundaries hold for that model, the two isotherms and the error models used; real columns add errors in geometry, dispersion and heat transfer that were held exact here. Observation through interior probes of uptake (O1) is a ceiling

that experiments rarely reach. The identifiability criterion asks whether the 95 % profile interval lies within a factor of two of k; it is a criterion of precision and accuracy, not of calibration. In an exploratory check the intervals contained the true k in only about 10 % of rows, because they are narrower than the small biases from time resolution and from errors in the inputs to q^*_{meas}. At the two highest Da levels the time resolution also biased the estimate itself, by factors of 0.6 to 24, so identifiability there is partly a property of the sampling; H2b does not depend on those levels. Symbolic regression ran on a subgrid, the neural pipeline only as a demonstration, and a Kolmogorov–Arnold network only in an exploratory run on its declared subgrid, where it recovered the law in none of 216 fits; each limit was set by cost rules declared before the grid ran. Finally, three errors in our own code and one solver setting that let a solve run for days were found and corrected during the study; the corrections, and the verdicts as first computed, are public (Supplementary Note 2), and none changed the main verdict.

Two directions follow. Discovery would gain most from better isotherms: the boundaries moved more with isotherm error than with noise, so an isotherm measured to a fraction of a per cent is worth more to discovery than quieter sensors. And the map should be tested where it is most uncertain, on experimental breakthrough data with interior probes at moderate Da, where it predicts that discovery should first become possible.

Methods

Ground truth

Ground truth came from the finite-difference reference solver of the companion study [26]: a one-dimensional, axially dispersed, non-isothermal model of a packed column, verified there against closed-form solutions. Uptake followed one of two laws,

$$\text{L1:}\quad \frac{\partial q}{\partial t} = k\,(q^*(c,T) - q), \tag{1}$$

$$\text{L2:}\quad \frac{\partial q}{\partial t} = k(q)\,(q^*(c,T) - q), \qquad k(q) = k_0\Big[1 - 0.8\,s\Big(\frac{q/q_{\max} - \theta}{0.05}\Big)\Big], \tag{2}$$

where s is the logistic function and θ is the fractional loading at the isotherm's step midpoint, so that under L2 the rate falls to 20 % of k_0 across the step. Two adsorbents were simulated with the same column geometry and flow (Supplementary Table 1): the companion study's favourable Langmuir sorbent, and water on a MOF-303-like framework, a cooperative (Type V) isotherm with its step at $P/P_0 = 0.15$, fed at 30 % relative humidity and 30 °C. The Damköhler number $\mathrm{Da} = k\,t_{\mathrm{stoich}}$ counts kinetic time constants per passage of the adsorption front, with t_{stoich} the stoichiometric time of the column (7200 s for Langmuir and 31 050 s for the step isotherm). Nine levels from 10^{-1} to 10^{3} at half-decade spacing therefore span k from 1.39×10^{-5} to $0.139\,\mathrm{s}^{-1}$ for Langmuir and from 3.22×10^{-6} to $0.0322\,\mathrm{s}^{-1}$ for the step isotherm. Three axial-dispersion coefficients, $D_L \times 0.2$, $\times 1$ and $\times 5$ of the default, completed the grid. They give column Péclet numbers $\mathrm{Pe} = uL/D_L$ of about 470, 94 and 19, and the levels are named by these values below (the code and the pre-registration

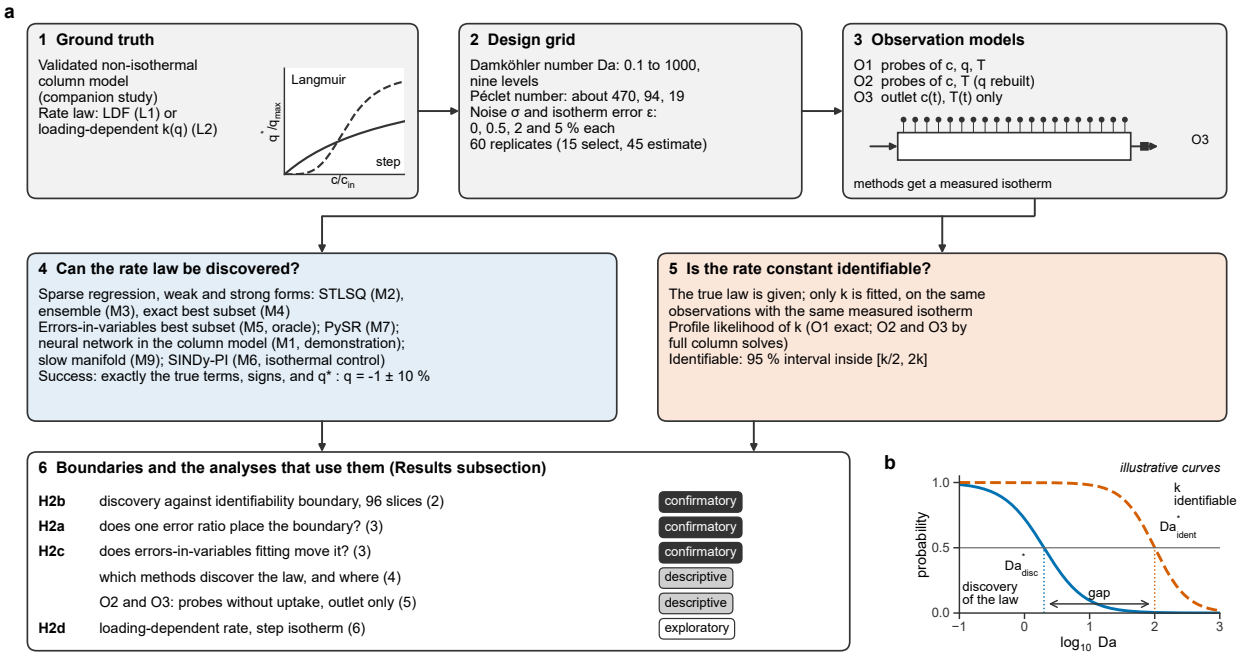


Figure 1: **Study design.** **a**, Breakthrough data are simulated by a validated column model for a Langmuir sorbent and a MOF-303-like step isotherm (box 1; isotherm shapes drawn from the model), over the design grid (box 2), and observed through interior probes with uptake (O1), probes without uptake (O2) or the outlet alone (O3) (box 3). The same observations, with the same measured isotherm, go to the discovery methods (box 4) and to a profile-likelihood test of k with the true law given (box 5). Box 6 lists the analyses with the Results subsection that reports each and whether it was confirmatory, exploratory or descriptive. **b**, What H2b compares in each slice: the Damköhler number at which the best discovery method falls to 50 % success ($\mathrm{Da}^*_{\mathrm{disc}}$) and the one at which k stops being identifiable in half the replicates ($\mathrm{Da}^*_{\mathrm{ident}}$). The curves in **b** are illustrative.

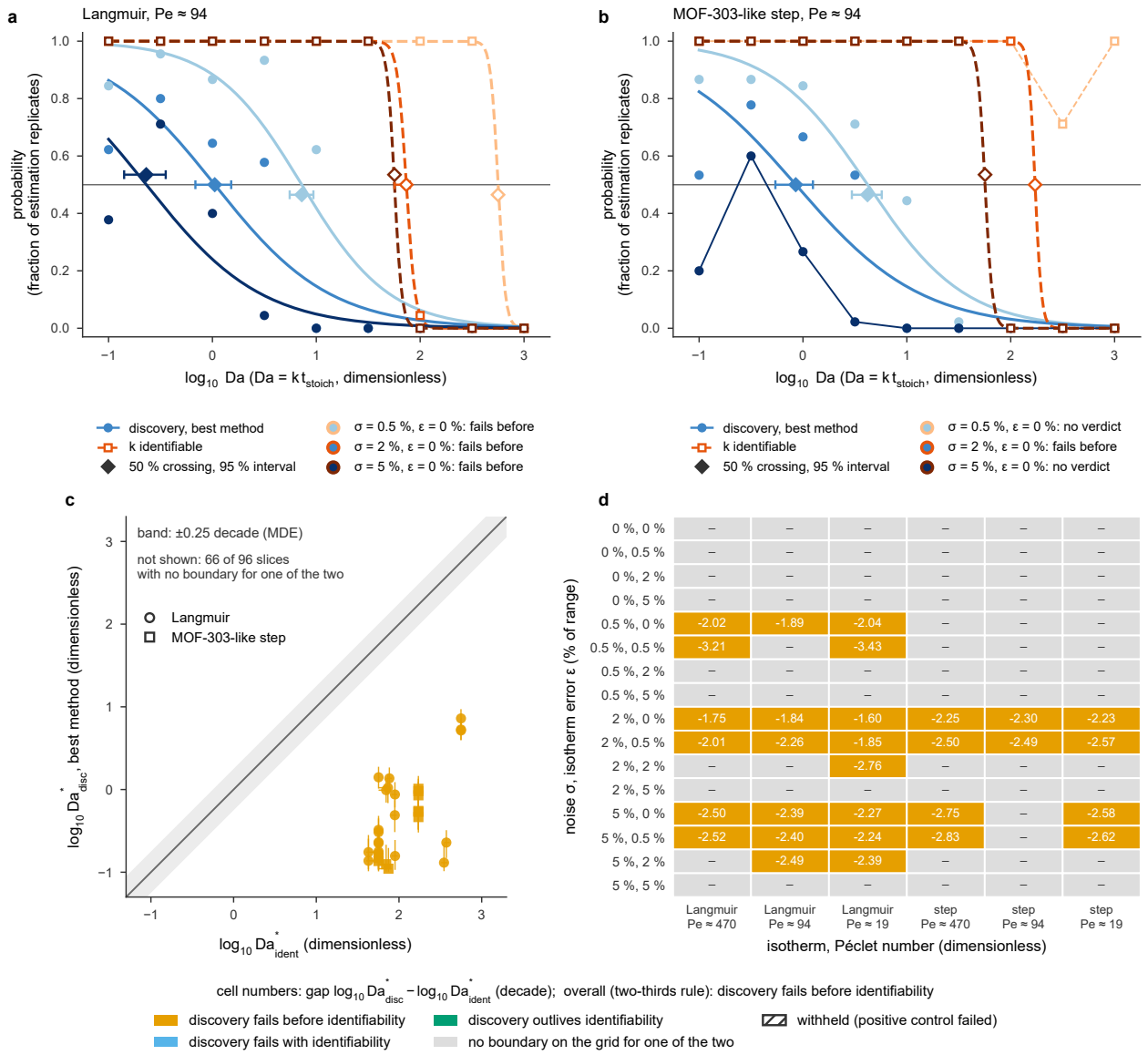


Figure 2: **Discovery fails long before identifiability.** **a**,**b**, Fraction of the 45 estimation replicates in which the best discovery method recovered the LDF law (filled circles, blue) and in which k was practically identifiable (open squares, orange), against Da, for an exact isotherm and noise of 0.5, 2 and 5 % (light to dark), at Pe ≈ 94; **a**, Langmuir sorbent; **b**, MOF-303-like step isotherm. Curves are the logistic fits that define the boundaries, drawn only where a 50 % crossing exists; otherwise the measured fractions are joined. Diamonds mark the crossings with 95 % bootstrap intervals. **c**, $\mathrm{Da}^{*}_{\mathrm{disc}}$ against $\mathrm{Da}^{*}_{\mathrm{ident}}$ for the 30 slices with both boundaries; the grey band is the pre-registered verdict threshold, the minimum detectable effect (±0.25 decade), around equality. **d**, The H2b verdict of every slice (rows: noise and isotherm error; columns: isotherm and Péclet number) with the gap $\log_{10}\mathrm{Da}^{*}_{\mathrm{disc}} - \log_{10}\mathrm{Da}^{*}_{\mathrm{ident}}$ in decades. Grey cells have no boundary on the grid for one of the two.

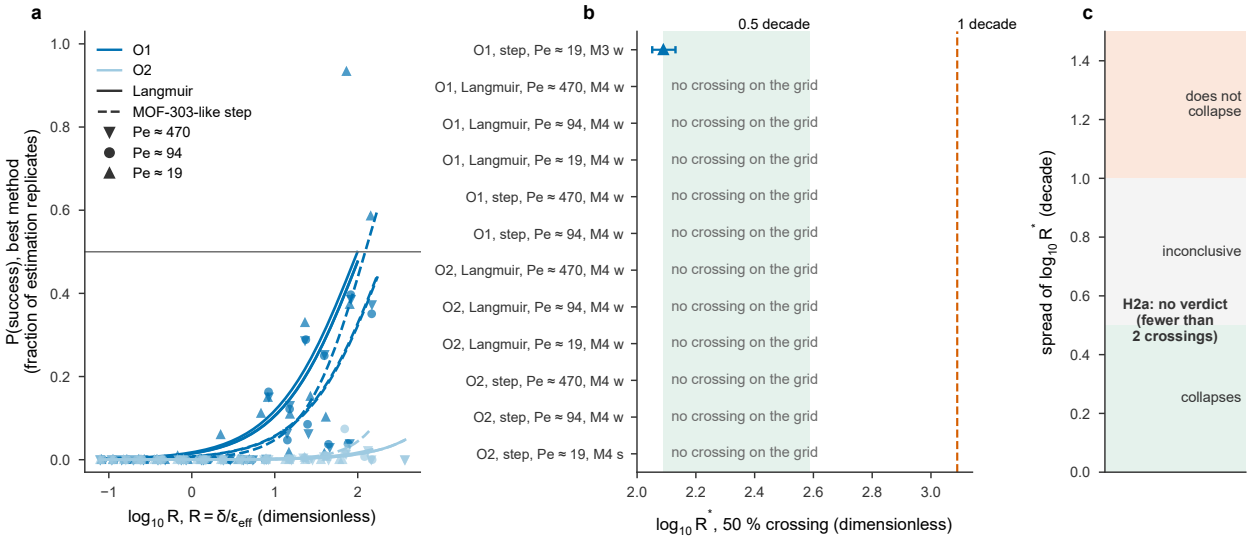


Figure 3: **No single error ratio places the discovery boundary (H2a).** **a**, Success of the best method in each condition against $R = \delta/\varepsilon_{\text{eff}}$, the realised driving force over the combined observation and isotherm error, on probes with uptake (O1, blue) and without (O2, orange). **b**, The 50 % crossing R^* of each condition with its 95 % interval; only one of the twelve conditions crosses. **c**, The pre-registered decision regions for the spread of $\log_{10} R^*$ across conditions; with fewer than two crossings, H2a has no verdict.

Table 1: **Pre-registered hypotheses and their verdicts.** Words in quotation marks are the pre-registered verdict words. Intervals are 95 % bootstrap intervals over the estimation replicates.

	Question	Result	Verdict
Positive control	Do discovery and identifiability both succeed at the lowest Da, without noise or isotherm error (≥ 90 %)?	Langmuir 135/135 and 135/135; step 135/135 and 135/135	Both isotherms enter every verdict
H2b	Does discovery fail before, with, or after identifiability (± 0.25 decade)?	30 of 96 slices with both boundaries; gap -1.60 to -3.43 decades (median -2.26 Langmuir, -2.54 step)	"*discovery fails before identifiability*" (30 of 30 slices)
H2b, k-aware	As H2b, with success also requiring $\hat{k}$ within a factor of 2	Unchanged	"*discovery fails before identifiability*"
H2a	Is the boundary at one value of $R = \delta/\varepsilon_{\text{eff}}$ (spread < 0.5 decade)?	One of 12 conditions crosses ($R^* = 123$, 113 to 135)	"*no verdict: fewer than two conditions have a crossing*"
H2c	Does errors-in-variables fitting move the boundary by $\geq 2\times$ at $\varepsilon > 0$?	No slice with a boundary for both estimators	"*no verdict*" at every ε
H2d (exploratory)	Is a loading-dependent rate discovered for the step isotherm?	At Da = 1 in 77 to 100 % of replicates (best method); at most 58 % elsewhere	Descriptive

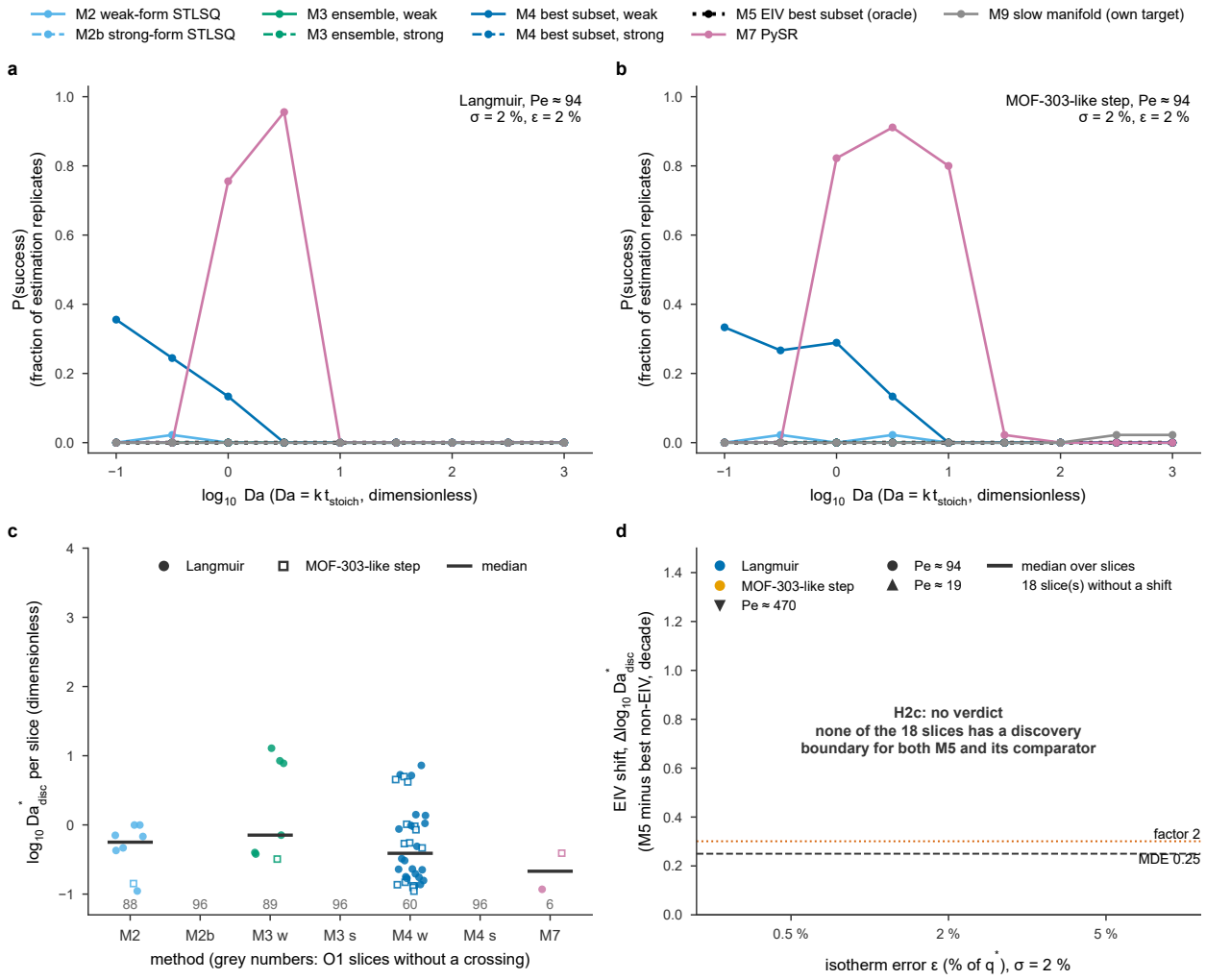


Figure 4: **Methods compared.** **a**,**b**, Success of every method against Da at 2 % noise and 2 % isotherm error, Pe ≈ 94; **a**, Langmuir; **b**, step isotherm. Symbolic regression (M7, orange) succeeds only in a window of intermediate Da, where every sparse method stays below 50 %. **c**, $\log_{10} \mathrm{Da}^{*}_{\mathrm{disc}}$ of each method in every slice where it crosses 50 %; the grey numbers count the slices without a crossing. **d**, H2c, the shift of the discovery boundary by the errors-in-variables estimator (M5) relative to the best strong-form comparator; no slice has a boundary for both, so H2c has no verdict.

label them by the D_L multiplier, so its "×0.2" level is the highest Pe). The grid has 81 solves at 400 cells and 400 stored times, each run until the outlet reached 95 % breakthrough, plus 9 isothermal solves for a control described below. The manifest that records every parameter is public.

Observation models, noise and isotherm error

Every method and every identifiability calculation read the data through one function, so all saw the same corrupted observations. Three observation models were used: interior probes of c, q and T (O1), interior probes of c and T with q hidden (O2), and the outlet $c(t)$ and $T(t)$ only (O3). O1 and O2 used 20 equally spaced probes; all models used 200 times spread evenly over the solved horizon. Noise was independent Gaussian with standard deviation $\sigma \in \{0, 0.5, 2, 5\}$ % of each channel's range. Methods were never given the true isotherm. They were given a measured one, $q^*_{\mathrm{meas}}(c,T) = q^*(c,T)\,[1 + \varepsilon\, g(c/c_{\mathrm{in}})]$ with $\varepsilon \in \{0, 0.5, 2, 5\}$ %, where g is a smooth random function of four Fourier modes on $[0,1]$, scaled to unit root-mean-square and drawn once per replicate. Every random draw was seeded by a hash: the noise by (cell, replicate, σ, observation model) and the isotherm error by (cell, replicate), so each observation can be regenerated on its own and every observation model sees the same measured isotherm. A replicate, one such observation of one cell, was the unit of analysis. Discovery on O1 and O2 and identifiability on O1 used 60 replicates per cell; the costlier O2 and O3 identifiability profiles used 3, and the M1 demonstration 1.

Discovery methods

Each method ran in the strongest configuration we could verify on known-answer systems before the pre-registration was frozen, and a method that could not be run that way was reported as such rather than replaced by a weaker one. The candidate library was $\{1, c, q, T, cq, q^2, c^2, q^*_{\mathrm{meas}}\}$; under L1 the true law is $k(q^*_{\mathrm{meas}} - q)$. The methods were sparse regression on a Savitzky–Golay estimate of $\mathrm{d}q/\mathrm{d}t$ (strong form) and on its weak, integral form with compactly supported test functions [3, 27], each with three solvers: sequentially thresholded least squares [1], a bagged ensemble with an inclusion threshold of 0.6 [5], and exact best-subset selection by enumeration (M4; exact subset selection as in [6], here scored by the Bayesian information criterion); errors-in-variables best-subset regression given the true error levels (M5, an oracle); implicit sparse regression without any isotherm on an isothermal control (M6, SINDy-PI [28]); symbolic regression (M7, PySR [7]); a universal differential equation with a neural network in the uptake term, trained through the column model and read out by sparse and symbolic regression (M1, the pipeline of [9]); and regression of the state on q-free features, which returns the isotherm where uptake is at equilibrium (M9 [18]). On O2, uptake was first recovered from the gas-phase mass balance, and the three sparse solvers were then run on it in both forms; M5, M7 and M9 were not run on O2, as the pre-registered applicability table states. 2 h per fit on the machine available, so it ran as a declared demonstration: Langmuir, Pe ≈ 94, $\sigma = 2$ %, $\varepsilon = 0$, O3, the nine Da levels, one replicate. The best parameters seen during training were kept. As first run, a column solve that exhausted its step budget during training ended the fit with no readout (four of the nine fits); the corrected code keeps the best parameters in that case,

and both runs are reported (Supplementary Note 2). A Kolmogorov–Arnold network [29] failed its known-answer test and its cost gate before the freeze and was not run on the grid. Method settings, solver thresholds and their known-answer tests are listed in Supplementary Note 1.

Success

Discovery succeeded when the selected terms were exactly those of the true law, every sign was correct and, for L1, the ratio of the q^*_{meas} and q coefficients was -1 within 10 %. The value of k itself was not required. A method that raised an error counted as a failure; no replicate, cell or output was excluded.

Identifiability

Practical identifiability of k was judged by profile likelihood [25] with the true law known, on the same observations and with the same measured isotherm as the discovery methods: k was practically identifiable when the 95 % profile interval (likelihood-ratio threshold 3.84) lay within $[k/2, 2k]$. On O1 the likelihood summed over all probes the fit of the known law, integrated exactly for piecewise-linear $q^*(t)$ along the observed c and T. On O2 and O3 every point of the profile was a full column solve. The profile searched a coarse logarithmic grid of k and refined it around the interval (41 points per pass on O1, 21 on O2 and O3, as pre-registered). Two corrections to the identifiability profiles were made during the study and are reported in full (Supplementary Note 2). First, on O2 and O3 a measured isotherm with $\varepsilon > 0$ could stall the stiff solver at large trial k for days; each solve was therefore capped at 600 s and a capped solve was scored as a failed solve (amendment A5; two O3 rows had capped solves, none next to its interval). Second, the O1 grid first spanned 10^{-5} to $10\,\mathrm{s}^{-1}$, which excluded the step cells at Da = 0.1; it was widened to 10^{-7} to $10\,\mathrm{s}^{-1}$, a peak on the grid edge was made an error, and all O1 rows were recomputed (amendment A6). Before that recomputation, the O1 refinement was found to use 21 points per pass instead of the pre-registered 41 and was set to 41, so every corrected O1 row uses the declared settings.

Boundaries and verdicts

For each slice the boundary Da^* was the 50 % crossing of an unpenalised logistic regression of replicate outcomes on $\log_{10}\mathrm{Da}$, with a percentile bootstrap interval (2000 resamples within each Da level). No boundary was reported, and no verdict issued, when the fitted curve did not cross 50 % inside the tested range or rose with Da instead of falling; the range was reported instead. Before the freeze the estimator's coverage on planted boundaries was 0.94 for a nominal 0.95, with a bias of +0.012 decade. The best discovery method of each slice was chosen on replicates 0–14 and its boundary estimated on replicates 15–59, which removes the inflation that choosing and estimating on the same replicates causes (0.07 decade in simulation); the identifiability boundary was estimated on the same 45 replicates. The best method was the one with the largest crossing on the selection replicates, or, if none crossed, the one with the highest mean success. Only rate-law methods that

ran on every cell of the slice competed: M5, which knows the true error levels, and M9, which targets the isotherm, were excluded.

Three hypotheses were confirmatory, each with verdict words fixed in advance. H2a: success collapses onto one curve in $R = \delta/\varepsilon_{\mathrm{eff}}$, the driving force relative to the combined error, with the spread of R^* across conditions under half a decade. H2b: on O1, for each slice (σ, ε, isotherm, Pe; 96 in all), the gap $\log_{10}\mathrm{Da}^*_{\mathrm{disc}} - \log_{10}\mathrm{Da}^*_{\mathrm{ident}}$ was called "*discovery fails before identifiability*" at -0.25 decade or less, "*discovery outlives identifiability*" at $+0.25$ or more, and "*discovery fails with identifiability*" in between; the hypothesis-level verdict was the word held by at least two thirds of the slices with a verdict. H2c: errors-in-variables regression moves the discovery boundary upwards by at least 0.25 decade in at least two thirds of the slices at a given $\varepsilon > 0$. With 45 estimation replicates the H2b rule returned a wrong directional verdict with probability 0.001 when there was no gap, and the correct one with probability 0.965 at a gap of 0.40 decade, the smallest effect of interest. A positive control guarded every verdict: in each isotherm's easiest cells (lowest Da, $\sigma = \varepsilon = 0$, O1, the three Péclet numbers pooled) the best method had to succeed and k had to be identifiable in at least 90 % of the estimation replicates, or that isotherm's verdicts were withheld. O2 and O3 were reported descriptively, and H2d, whether the loading-dependent rate of L2 can be discovered, was exploratory.

Pre-registration and amendments

The design was frozen on 11 October 2026 (commit `7dcc2fe`) before any discovery method ran on any grid cell, and every confirmatory driver refused to run unless the pre-registration matched its latest freeze commit. (The study's history was moved from the companion project's repository into its own; the pre-registration's amendment headers cite the original commit identifiers, and a published table maps them to the ones given here.) Six dated amendments are appended to it. A1 to A4 were made before the grid ran: 60 replicates split into selection and estimation sets, the smallest effect of interest, the oracle kept out of the pooled verdicts, Pe as a slicing axis, errors counted as failures, the decision to run the grid before peer review, and the M1 demonstration. A5 (the solve cap) was made while the grid ran, and A6 (the O1 grid range) after the verdicts were first computed. Both as-run and corrected verdicts are public, and the paper reports both (Supplementary Note 2).

Exploratory analyses

Three analyses were declared in writing before they ran and are reported as exploratory: M1 trained at the scale a universal differential equation needs (E1), the Kolmogorov–Arnold network on its declared subgrid once the machine was free (E2), and how often an interval that meets $[k/2, 2k]$ misses the true k (E3); both are reported in Supplementary Note 3. E3 used the corrected identifiability rows. E1 needs a graphics processor and has not run; it is not reported here. Three further descriptive checks were made after the verdicts were seen and are labelled so: the accuracy

of the O1 profile with exact data at each Da, H2b recomputed without the two highest Da levels, and a classification of the slices without a verdict by which boundary is missing.

Boundaries with perfect separation

Where every replicate is identifiable at one level of Da and none at the next, the unpenalised logistic fit has no finite maximum-likelihood slope: the optimiser stops at a steep curve whose crossing lies between the two levels, near their geometric midpoint. Every bootstrap resample reproduces the same data, so the interval has zero width. Such a boundary is reported as lying between those two levels; the gap used by H2b is computed from it exactly as from any other crossing.

Software and computation

The grid ran on a single computer without a graphics processor, one job at a time, from 12 October to 1 November 2026; the corrected O1 identifiability rows and the four M1 refits were computed from 1 to 4 November. Package versions are pinned in the repository.

Use of AI tools

During the preparation of this work the author used a generative AI assistant (Claude, Anthropic) as a tool to assist with drafting and language editing, with writing and checking code, and with checking bibliographic records. The author designed the study, made every scientific decision, reviewed and edited all content, and takes full responsibility for the content of the published article.

Data availability

The ground-truth manifest, every results file, the run log and the as-run and corrected verdicts are public in the study repository and archived on Zenodo (doi:10.5281/zenodo.23325994). The ground-truth fields are regenerated from the manifest by `p2_generate.py`.

Code availability

All code is public at https://github.com/abdulhamidbatayhi123/ai-kinetics-discoverability under the MIT licence and archived with the data. The solver is imported unchanged from the companion study's archive (doi:10.5281/zenodo.23323685).

Author contributions

A.B. is the sole author and conceived and designed the study. The use of AI tools is described in Methods.

Competing interests

The author declares no competing interests.

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